

MCR-inhibitor baseline — ADMET-AI vs Admetica

A reference baseline: both predictors run on the three best-characterised methyl-coenzyme M reductase (MCR) inhibitors, so every endpoint in this repo has a known-inhibitor anchor to compare Anna's candidates against.

Inhibitor	SMILES	role	MW
3-NOP (3-nitrooxypropan-1-ol)	<chem>`O=[N+]([O-])OCCCO`</chem>	the feed-additive MCR inhibitor actually in commercial use	121.09
coenzyme M (2-mercaptoethanesulfonic acid)	<chem>`O=S(=O)(O)CCS`</chem>	the native MCR substrate/product; surrogate inhibitor reference	142.20
rosmarinic acid	<chem>`O=C(/C=C/c1ccc(O)c(O)c1)OC(Cc1ccc(O)c(O)c1)C(=O)O`</chem>	plant-derived MCR inhibitor	360.32

All three SMILES were RDKit-validated (canonical SMILES + formula + MW match the known values: C3H7NO4 / C2H6O3S2 / C18H16O8). Run with:

```
```sh .venv/bin/python code/admet_ai_client.py --input mcr_inhibitors.csv --no-percentiles --no-physchem -o mcr_inhibitor_baseline_admet_ai.csv .venv/bin/python code/admetica_client.py --input mcr_inhibitors.csv -o mcr_inhibitor_baseline_admetica.csv ```
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ADMET-AI = 41 ADMET endpoints (classification = prob [0,1]; regression = native units). Admetica = 22 endpoints + 15 applicability-domain (AD) scores.

## 1. ADMET-AI baseline (41 endpoints)

endpoint	3-NOP	coenzyme M	rosmarinic acid	notes
AMES	0.26	0.48	0.37	none flagged (all <0.5)
BBB_Martins	0.94	0.83	0.06	small polar pair predicted high BBB; rosmarinic acid excluded from brain (large, polar)
Bioavailability_Ma	0.91	0.62	0.35	3-NOP high; rosmarinic acid low oral F
Carcinogens_Lagunin	0.48	0.61	0.15	none flagged
ClinTox	0.03	0.09	0.17	none flagged
DILI	0.39	0.18	0.55	rosmarinic acid just over 0.5 (mild DILI signal)
HIA_Hou	1.00	0.67	0.59	3-NOP fully absorbed
hERG	0.017	0.015	0.16	all green — no cardiotoxicity signal from ADMET-AI
PAMPA_NCATS	0.79	0.17	0.06	3-NOP permeable; rosmarinic acid not
Pgp_Broccatelli	0.001	0.001	0.007	none are Pgp substrates per ADMET-AI
Skin_Reaction	0.71	0.72	0.71	all flagged skin sensitization
CYP1A2_Veith	0.026	0.002	0.17	none inhibit
CYP2C19_Veith	0.022	0.008	0.070	none inhibit
CYP2C9_Veith	0.004	0.004	0.137	none inhibit
CYP2D6_Veith	0.004	0.006	0.064	none inhibit
CYP3A4_Veith	0.0002	0.0002	0.151	none inhibit
CYP2C9_Sub	0.23	0.11	0.045	weak; none strong substrates
CYP2D6_Sub	0.076	0.031	0.009	none
CYP3A4_Sub	0.29	0.14	0.18	weak; none strong substrates

endpoint	3-NOP	coenzyme M	rosmarinic acid	notes
NR-* (12 assays)	all <0.23	all <0.05	0.06–0.23	no endocrine/receptor flags on any of the three
SR-* (5 stress response)	all <0.05	all <0.25	0.04–0.39	no genotoxic/oxidative-stress flags
Caco2_Wang	–4.19	–4.91	–6.31	log-scale; low permeability for all (rosmarinic acid lowest)
Clearance_Hepatocyte_AZ	59.5	9.0	38	3-NOP fastest hepatic clearance
Clearance_Microsome_AZ	8.7	–20.1	27	coenzyme M predicted near-zero microsomal clearance
Half_Life_Obach	–4.8	14.7	–25.8	regression (log scale); values unreliable-looking for extremes
HydrationFreeEnergy_FreeSolv	–7.0	–10.8	–22.1	rosmarinic acid most hydrophilic
LD50_Zhu	1.92	1.80	1.99	all similar, low acute toxicity (mol/kg scale)
Lipophilicity_AZ	–0.02	–0.96	–0.94	all hydrophilic
PPBR_AZ	28.2	36.6	85	rosmarinic acid strongly protein-bound (polyphenol)
Solubility_AqSolDB	–0.48	–0.31	–2.55	rosmarinic acid least soluble
VDss_Lombardo	–1.94	–0.11	2.33	rosmarinic acid highest distribution volume

**Read:** ADMET-AI reads all three as clean on the safety-critical endpoints — no hERG, no AMES, no endocrine-receptor, no CYP inhibition, low acute toxicity. The only ADMET-AI reds are **skin sensitization on all three** (the small nitrate/thiol/sulfonate groups are alert-bearing) and a mild **DILI signal on rosmarinic acid** (0.55). 3-NOP and coenzyme M read as small, well-absorbed, BBB-permeable, hydrophilic molecules; rosmarinic acid reads as a larger, polar, low-oral-F, high-protein-binding polyphenol that stays out of the brain.

## 2. Admetica baseline (22 endpoints + AD)

### 2a. Endpoint values

endpoint	3-NOP	coenzyme M	rosmarinic acid	notes
Caco2	–5.4	–4.85	–7.13	agrees with ADMET-AI ordering (rosmarinic acid lowest)
Lipophilicity	0.35	–2.18	0.64	3-NOP +0.35 vs ADMET-AI –0.02 — <b>disagrees</b> (low AD)
Solubility	–0.27	0.75	–2.0	agrees on rosmarinic acid (least soluble); coenzyme M sign differs
Pgp-Inhibitor	~0	~0	0.0005	none inhibit Pgp
Pgp-Substrate	~0	~0	0.90	<b>rosmarinic acid is a strong Pgp substrate</b> (ADMET-AI missed — it doesn't model this)
PPBR	44	–35	92	rosmarinic acid high (agrees); coenzyme M –35 is <b>nonsense</b> (AD 0.23, out of domain)
VDss	32.6	3.49	–0.22	sign differs from ADMET-AI (different dataset/units — rank only)
CYP1A2-Inhib	~0	~0	0.008	none inhibit (agrees with ADMET-AI)
CYP1A2-Sub	0.43	0.041	0.056	3-NOP moderate CYP1A2 substrate
CYP2C19-Inhib	0.91	~0	0.0006	<b>3-NOP flagged CYP2C19 inhibitor</b> (ADMET-AI said 0.02) — AD 0.24, low
CYP2C19-Sub	0.002	0.008	~0	none
CYP2C9-Inhib	0.35	~0	0.071	3-NOP moderate (ADMET-AI 0.004) — low AD
CYP2C9-Sub	0.032	0.52	0.94	<b>CYP2C9-substrate systematic miscalibration</b> (see addendum §3) — Admetica calls rosmarinic acid + coenzyme M substrates, ADMET-AI says no

endpoint	3-NOP	coenzyme M	rosmarinic acid	notes
CYP2D6-Inhib	~0	~0	0.003	none
CYP2D6-Sub	0.023	0.0001	0.008	none
CYP3A4-Inhib	~0	~0	0.091	none
CYP3A4-Sub	0.69	0.010	0.96	3-NOP + rosmarinic acid strong CYP3A4 substrates
CL-Hepa	18.6	16.7	40.6	agrees ordering with ADMET-AI (rosmarinic acid highest)
CL-Micro	77.9	−37.1	21.8	coenzyme M negative = out-of-domain nonsense (AD 0.25)
Half-Life	−3.3	223	0.70	coenzyme M 223 is out-of-domain nonsense (AD 0.24)
hERG	0.0005	<b>0.746</b>	0.133	<b>coenzyme M flagged hERG binder by Admetica</b> — but hERG has no AD, and ADMET-AI gives 0.015. Un-arbitratable; treat as caution
LD50	2.86	2.67	1.50	same scale as ADMET-AI; rosmarinic acid (lower LD50) agrees as more toxic

2b. Applicability-domain (AD) scores

endpoint	3-NOP	coenzyme M	rosmarinic acid
Caco2	0.24	0.21	0.43
Lipophilicity	0.22	0.23	0.40
Solubility	0.34	0.28	0.46
Pgp-Substrate	0.25	0.22	0.42
PPBR	0.23	0.23	0.40
VDss	0.29	0.24	0.42
CYP1A2-Inhib	0.24	0.23	0.41
CYP2C19-Inhib	0.24	0.24	0.41
CYP2C9-Inhib	0.24	0.24	0.41
CYP2C9-Sub	0.25	0.22	0.42
CYP2D6-Inhib	0.24	0.24	0.42
CYP2D6-Sub	0.24	0.21	0.41
CL-Hepa	0.23	0.24	0.42
CL-Micro	0.24	0.25	0.42
Half-Life	0.26	0.24	0.42

**Domain read: 3-NOP and coenzyme M are far out of Admetica's training domain** (AD 0.21–0.34 across every endpoint) — they are tiny, polar, heteroatom-dense molecules unlike most drug-like training sets. Their Admetica regression calls (PPBR −35, CL-Micro −37, Half-Life 223 for coenzyme M; Lipophilicity +0.35 for 3-NOP) are therefore **unreliable and should be discounted**. **Rosmarinic acid is in-domain** (AD 0.40–0.46) — its Admetica calls are the trustworthy ones here.

3. What the baseline tells us for Anna's candidates

1. **Both models agree the known inhibitors are ADMET-clean** on the decision-critical endpoints (hERG, AMES, endocrine receptors, CYP inhibition, acute toxicity). This is reassuring: the \*positive controls\* don't light up as toxic, so the candidates that do flag on those endpoints are flagging for chemistry, not because MCR-inhibitor chemistry is generically flagged. 2. **Skin sensitization is expected** for this chemistry (all three inhibitors are flagged) — consistent with the addendum treating skin reds as a handling note,

not a deprioritisation driver. 3. **The hERG disagreement is real here too:** Admetica flags **coenzyme M** hERG 0.75 (ADMET-AI 0.015, no AD) — same over-aggressive hERG pattern seen in Anna's set. Reinforces the addendum's "hERG is the contested endpoint" conclusion. 4. **Admetica's CYP2C9-substrate and substrate calls are miscalibrated here too** (rosmarinic acid called a 0.94 CYP2C9 substrate; ADMET-AI says 0.045) — the same systematic offset noted in the addendum, now on a known molecule. 5. **The AD score does its job:** it correctly identifies 3-NOP and coenzyme M as out-of-domain, and the obviously-nonsensical Admetica regression outputs (negative clearances, half-life 223) all sit on the low-AD rows. This is the AD score earning its place as the second opinion's confidence gauge. 6. **3-NOP, the actual commercial MCR inhibitor, reads as a clean, small, well-absorbed, non-Pgp-substrate molecule** under ADMET-AI — a sanity check that the modelling stack lands in the right place for the one compound we know is field-viable.